Project Initialization and Planning Phase

Date	10/07/2026
Name	Jaswanth Madapuram
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report:

The proposed solution aims to automate the credit card approval process. The system analyzes applicant information such as annual income, employment duration, age, education level, payment history, credit inquiries, and other financial details to predict whether a credit card application should be approved or rejected. This solution reduces manual effort, improves prediction accuracy, minimizes processing time, and supports faster decision-making for financial institutions.

Project Overview	
Objective	To develop a machine learning-based system that predicts credit card approval accurately by analyzing applicant financial and personal information, reducing manual verification time and improving decision-making.
Scope	The project focuses on collecting applicant details, preprocessing the data, training machine learning models, predicting approval status, and displaying the final decision through a web application.
Problem Statement	
Description	Traditional credit card approval is time-consuming and depends heavily on manual verification, increasing the chances of errors, delays, and inconsistent decisions.
Impact	Automating the approval process improves operational efficiency, reduces processing time, minimizes financial risks, and ensures consistent and accurate credit decisions.
Proposed Solution	

Approach	Build a supervised Machine Learning model using historical applicant data. The system preprocesses applicant information, extracts relevant features, predicts approval status, and displays the result through a Flask-based web application.
Key Features	• Machine Learning-based credit card approval prediction. • Automated applicant data validation and preprocessing. • Prediction using trained classification algorithms. • Fast and accurate approval/rejection decisions. • User-friendly web interface developed using Flask. • Model persistence using Joblib for quick predictions. • Data visualization for approval trends and applicant analysis. • Scalable architecture for future integration with banking systems.

Resource Requirements:

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5/i7 Processor (or equivalent)
Memory	RAM	8 GB RAM (Minimum)
Storage	Disk space for dataset, trained model, and application	20 GB Free Storage (SSD Preferred)
Software		
Framework	Web Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		
Dataset	Source, size, format	Credit Card Approval Dataset (CSV format), approximately 200+ records used for training and testing